

Fraction word problems (remainder trap + multiple steps)

Unit three · fractionapplicationquestion · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5A Unit 3 + Modern Education Volume 5A Unit 11-12

Core Trap: ☐ T3 "Remaining Fraction" Baseline Quantity Change + T7 Multi-step Reasoning Chain

SSPA Related: ● High Frequency The two-word question in the score paper is compulsory, accounting for about 25-30% of the application questions

Prerequisite knowledge: Class 8 (Addition of different denominators), Class 9 (Multiplication of fractions - cross reduction method), application questions of addition and subtraction of fractions

Goal of this class: ❶ Fraction "is what fraction of..." ❷ Remaining trap (change in baseline quantity) ❸ Multi-step reasoning chain ❹ Reverse reasoning (finding the original quantity with a known result)

Student Name:

Class:

Date:

Time Spent:

I.Warm-Up Questions (total 5 question , 5 minutes)

#	Question	Difficulty	Working Space (Show full working)
1	Calculate $\frac{2}{3} \times \frac{1}{2} = ?$ (Cross reduction, Tang 9 basis)	Basic	
2	Calculate $1 - \frac{1}{4} = ?$ ("remaining" = 1 – used, P4 basis)	Basic	
3	A rope is 3 meters long and uses SEP . How many meters were used? ("Fraction of the total length" = multiplication) $\frac{1}{3}$. How many meters were used? ("Fraction of the total length" = multiplication)	Basic	
4	Xiao Ming has 12 candies. After eating SEP , how many are left? (Please tell me how many pills you have eaten first, and then ask for the rest) $\frac{1}{4}$, how many pills are left? (Please tell me how many pills you have eaten first, and then ask for the rest)	Basic	
5	Xiaomei has \$60. After using SEP , how many dollars are left? Then use the "remaining SEP ", how much more is used? (Note that the two reference amounts are different!) $\frac{1}{3}$, how many yuan are left? Then use the remaining $\frac{1}{4}$, how much more was used? (Note that the two reference amounts are different!)	Advanced	

II.Core Knowledge + Worked Examples

Knowledge point 1: Fraction problem - "What fraction of..." ● SSPA

Core formula:

① "C divided by B of A" = $A \times \frac{C}{B}$ (Keyword "of" = multiplication)

② "A is a fraction of B" = $A \div B = \frac{A}{B}$ (Comparative relationship)

③ Find "part" |||SEP|||: **Total amount $\times$ fraction = partial amount** Find "fraction" |||SEP|||: **Partial quantity $\div$ Total quantity = Fraction** (not in depth in this chapter)

④ Find the "percentage": **Part amount $\div$ total amount = fraction** (this chapter will not go into detail)

❏ Trap detonation example

There are 40 people in the class, and $\frac{3}{5}$ is a boy. How many boys are there?

✗ Common mistakes

$$40 \div \frac{3}{5} = 66.7 ?$$

✓ Correct solution

24 people

$$40 \times \frac{3}{5} = \frac{120}{5} = 24$$

"The" is multiplication, not division! 40×35 is right

 **Tip: "When you see "is", multiply, and when you see "yes", divide; the base quantity must be clearly seen, it may be different the first time and the second time."**

 **Key distinction: "C/B of A" $\rightarrow A \times C/B$ (multiplication). "A is what fraction of B" $\rightarrow A \div B$ (division).**

Knowledge Point - Worked Examples

#	Question	Difficulty	Working Space
Example 1	A bottle of water $\frac{1}{2}$ liters, Xiao Ming drank SEP . How many liters did you drink? $\frac{2}{3}$. How many liters did you drink?		
Example 2	A ribbon is $\frac{3}{4}$ meters long, cut off SEP . How many meters were cut off? $\frac{1}{2}$. How many meters were cut off?		

Knowledge point 2: "What fraction is left" - the benchmark quantity trap SSPA must-take killer questions

This is the trap that most students fall into!

① The essence of the trap |||SEP|||: After the first use, the "remaining" is no longer the original total ② The base amount of the second operation =

The remaining amount of the first time (not the original total!) Standard process

③ Step 1: Find the amount used for the first time (original total amount) $\times$ fraction ₁ :

Step 2: Find the amount left in the first time (original total amount – used)

Step 3: Find the amount used in the second time (amount left in the first time $\times$ fraction ₂)

Common but wrong |||SEP|||: directly use the original total amount $\times$ fraction ₂ (ignoring that the base amount has changed!)

④ Common but wrong: Directly use the original total amount $\times$ fraction ₂ (ignoring that the base amount has changed!)

Trap comparison questions (core of this class)

A rope is 24 meters long. Use $\frac{1}{3}$ first, and then use the remaining |||SEP|||. How many meters did it take the second time? $\frac{1}{2}$. How many meters did it take the second time?

Typical errors (80% students)

$$24 \times \frac{1}{2} = 12 \text{ meters}$$

Correct solution

8 meters

Step1: $24 \times \frac{1}{3} = 8$ (used) Step2: $24 - 8 = 16$ (remaining) Step3: $16 \times \frac{1}{2} = 8$ (used for the second time)

Just use the original total amount $\times \frac{1}{2}$, but the base amount has become 16 meters!

 **Tip: "Every time you use it, count how much is left. The base number will be different the next time you use it!"**

 **The deadliest trap: "The remaining fraction" - the "base amount" must be re-confirmed before each operation!**

Knowledge Point 2 Worked Examples

#	Question	Difficulty	Working Space
Example 3	There are 24 candies in a box. I ate $\frac{1}{3}$ on the first day and "the remaining SEP " on the second day. How many pills did you take the next day? (Note that the base amount has changed) $\frac{1}{2}$. How many pills did you take the next day? (Note that the base amount has changed)		
Example 4	A barrel of oil is 10 liters, use $\frac{2}{5}$ for the first time, and use the remaining $\frac{1}{3}$ for the second time. How many liters were used the second time?		

Knowledge point 3: Multi-step fraction word problems (continuous reasoning chain) SSPA Advanced

Continuous surplus problem - the baseline amount must be recalculated at each step:

- ① **Draw a flow chart** |||SEP|||: Total amount $\rightarrow$ First step (used) $\rightarrow$ Remaining $_1 \rightarrow$ Second step (used) $\rightarrow$ Remaining $_2 \rightarrow \dots$ Step-by-step calculation |||SEP|||: Do not skip steps, record "how much is used" and "how much is left" at each step
- ② **Verification skills** |||SEP|||: Add up the amount of each step + final remaining = total amount (check calculation): Don't skip steps, record "how much is used" and "how much is left" at each step
- ③ **Verification skills**: Sum up the amounts used in each step + final remaining = total amount (check calculation)

①

Draw a flow chart

Total $\rightarrow$ Use $_1 \rightarrow$ Left $_1 \rightarrow$ Use $_2 \rightarrow$ Left $_2 \rightarrow \dots$

②

Calculate step by step

Record used + remaining at each step

③

Confirm base amount

The base amount of points used each time is the remaining amount from the last time

④

Check calculation

Amount used in each step + final remaining amount = total amount

example

Example 5: Mom has \$100, and uses $\frac{1}{4}$ to buy vegetables, and the remaining $\frac{1}{3}$ to buy fruits. How much did it cost to buy fruit?

example

Example 6: A bottle of oil $\frac{2}{3}$ liters, use $\frac{1}{2}$ for the first time, and use the remaining $\frac{1}{3}$ for the second time. How many liters did you use the second time?

Knowledge Point 4: Reverse Reasoning - Find the Original Quantity from a Known Result ●
SSPA Paper 2 Killer Question

Reverse reasoning (reduction problem):

① Knowing the final remaining amount, find the original total amount

② **Key: working backwards!**

If $\frac{1}{3}$ is used, the remaining amount = total amount $\times (1 - \frac{1}{3}) = \text{total amount} \times \frac{2}{3}$

So: total amount = remaining amount $\div (1 - \text{fraction})$

③ **Simple formula |||SEP|||: original amount = remaining amount $\div$ remaining fraction:** Original amount = remaining amount $\div$ remaining fraction

example

Example 7: Xiao Ming has some sugar, and after using |||SEP|||, there are 12 pieces left. How many grains were there? (Hint: What percentage is left?) $\frac{1}{3}$ There are 12 pills left. How many grains were there? (Hint: What percentage is left?)

example

Example 8: A box of apples, after selling |||SEP|||, there are 30 apples left. How many were there? (Hint: remaining fraction = $1 - \frac{2}{5}$ 30 are left. How many were there? (Hint: remaining fraction = $1 - \frac{2}{5} = \frac{3}{5}$)

🧠 **Tip (reverse reasoning): "It doesn't matter how much you use, find out what percentage is left. Divide the remaining percentage by the low amount, and get back to the original!"**

III. Lesson layered synchronization practice (total 15 questions)

Basic layer (total 5 questions, everyone must do)

#	Question	Difficulty	Working Space
9	There are 36 people in the class, $\frac{1}{3}$ is short-sighted. How many people are nearsighted?		
10	A ribbon is $\frac{2}{3}$ meters long and uses SEP . How many meters were used? $\frac{1}{2}$. How many meters were used?		
11	There are 12 egg tarts in a box and I ate SEP . How many are left? $\frac{1}{3}$. How many are left?		
12	Dad' s monthly salary is \$30,000, and he uses $\frac{1}{5}$ to pay the rent. How much is the rent?		
13	A watermelon weighs 2 kg and is eaten SEP . How many kilograms did you eat? $\frac{3}{4}$. How many kilograms did you eat?		

Advanced layer (total 5 questions, choose do)

#	Question	Difficulty	Working Space
14	A barrel of oil is 5 liters, use $\frac{1}{5}$ on the first day, and use the "remaining SEP " on the second day. How many liters will be used the next day? $\frac{1}{4}$ ” . How many liters will be used the next day?		
15	A rope is 3 meters long. Use $\frac{1}{3}$ meters first, and then use the remaining $\frac{1}{2}$ meters. How many meters are left at the end?		
16	I have 20 kilograms of rice at home. I eat $\frac{1}{4}$ in the first week and "the remaining SEP " in the second week. How many kilograms did you eat in the second week? $\frac{1}{5}$ ” . How many kilograms did you eat in the second week?		
17	My sister had \$240 and used $\frac{1}{3}$ to buy books, and the remaining $\frac{1}{2}$ to buy stationery. How much did you spend on stationery?		
18	For a cake, my father ate SEP , my mother ate "the remaining SEP ", and my brother ate "the remaining SEP ". What fraction of the whole cake did my brother eat? $\frac{1}{4}$, mother ate "the remaining $\frac{1}{3}$ ” , the younger brother ate “and the rest $\frac{1}{2}$ ” . What fraction of the whole cake did my brother eat?		

challenge layer (total 5 questions, choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
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19	The warehouse has 180 bales of rice. Transport $\frac{1}{3}$ on the first day, transport "the remaining SEP " on the second day, transport "the remaining SEP " on the third day. How many packages were shipped on the third day? $\frac{1}{4}$, and on the third day the remaining ones were transported away $\frac{1}{2}$. How many packages were shipped on the third day?		
20	A book has 240 pages. I watched $\frac{1}{4}$ on the first day, "The rest of SEP " on the second day, and "The rest of SEP " on the third day. How many pages are left to read? $\frac{1}{3}$, and on the third day I watched "The rest $\frac{1}{2}$ " . How many pages are left to read?		
21	A rectangle is long $\frac{3}{4}$ meters and its width is long SEP . How many square meters is the area? (Find width first, then area) $\frac{2}{3}$. How many square meters is the area? (Find width first, then area)		
22	A bottle of juice contains $1\frac{1}{2}$ liter. The first time I poured SEP , the second time I poured "the remaining SEP ". How many liters are left? $\frac{1}{3}$, poured out for the second time "the rest $\frac{1}{2}$ " . How many liters are left?		
23	Xiaomei has some stickers. I gave $\frac{1}{4}$ to Xiao Hua, and then gave "the remaining SEP " to Xiao Ming, and finally there were 18 pieces left. How many stickers does Xiaomei have? (reverse reasoning) $\frac{1}{3}$ "To Xiao Ming, there are 18 cards left. How many stickers does Xiaomei have? (reverse reasoning)		

IV.fractionapplicationquestionspecial topic training (total 8 questions, all must do)

#	Question	Difficulty	Working Space
24	Residential buildings were built on $\frac{1}{4}$ on one site, and shopping malls were built on the remaining SEP . What fraction of the floor space does the mall occupy? (Note: the base amount is the "remainder" !) $\frac{2}{5}$ A shopping mall was built. What fraction of the floor space does the mall occupy? (Note: the base amount is the "remainder" !)		

25	The school has 800 students, $\frac{3}{8}$ are primary school students (P1-P3), and the remaining $\frac{2}{5}$ are middle school students (P4-P6). How many middle school students are there?		
26	One road is 2 kilometers long and has been repaired $\frac{3}{8}$. The remaining $\frac{1}{5}$ will be repaired next week. How many kilometers will be repaired next week?		
27	Garden $\frac{2}{5}$ grows flowers, flowers $\frac{3}{4}$ is roses. What percentage of the garden do roses occupy? (Two layers of "的" = continuous multiplication)		
28	There are 12 liters in a bucket of water. Use $\frac{1}{4}$ first, then add $\frac{1}{3}$ liters. How many liters are there now? (Mixed addition, subtraction and multiplication!)		
29	Xiao Ming had \$150, so he used $\frac{1}{5}$ to buy pencils, "the remaining $\frac{1}{3}$ " to buy books, and "the last remaining $\frac{1}{2}$ " to buy snacks. How much did you spend on snacks?		
30	A rectangular planter is $2\frac{1}{2}$ meters long and $\frac{1}{2}$ wide. How many square meters is the area? (Mixed fraction $\times$ fraction $\rightarrow$ turn to false first)		
31	Brother, I have some candy. After he ate $\frac{1}{5}$, he had 40 pills left. How many grains were there? (reverse reasoning) Afterwards, 40 pills remain. How many grains were there? (reverse reasoning)		

V. Ultimate challenge area (total 3 question)

#	Question	Difficulty	Working Space
 1	A book, read $\frac{1}{3}$ on the first day, and read "the remaining $\frac{3}{5}$ " on the second day, with 40 pages left. How many pages does this book have? (Ultimate reverse reasoning - two-level reduction)		
 2	The water tank contains 20 liters of water. Use $\frac{1}{5}$ on the first day, use "the remaining $\frac{1}{4}$ " on the second day, use "the remaining $\frac{1}{3}$ " on the third day, and use "the remaining $\frac{1}{2}$ " on the fourth day. How many liters were used on the fourth day? How many liters are left at the end? (Four-step reasoning chain)		

 3	Three children share a box of chocolates. A took SEP , B took "the remaining SEP ", C took "the remaining SEP ", and finally there were 6 pills left in the box. How many pills are in a box? (Triple reduction reasoning) $\frac{1}{4}$, B took "the remaining $\frac{1}{3}$", C took "the rest $\frac{1}{2}$", and finally there were 6 pills left in the box. How many pills are in a box? (Triple reduction reasoning)		
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VI.Class afterhomework

Basic must-do questions (total 5 questions)

#	Question	Difficulty	Working Space
H1	A rope is 5 meters long and uses SEP . How many meters were used? $\frac{1}{5}$. How many meters were used?		
H2	A bottle of juice $\frac{1}{2}$ liters, drank $\frac{1}{2}$ in it. How many liters did you drink?		
H3	There are 24 biscuits in a box and I ate SEP . How many pieces are left? $\frac{2}{3}$. How many pieces are left?		
H4	A potted flower is $\frac{1}{2}$ meters high. After watering, it grows taller than the original SEP . How many meters tall is it now? (Note: I have grown $\frac{1}{2}$ of my original height) $\frac{1}{2}$. How many meters tall is it now? (Note: I have grown $\frac{1}{2}$ of my original height)		
H5	My sister had \$120, so she used $\frac{1}{3}$ to buy books, and then used the "remaining SEP " to buy handmade materials. How much did it cost to buy handmade materials? $\frac{1}{4}$ Buy handmade materials. How much did it cost to buy handmade materials?		

Advanced choose doquestion (total 3 questions, choose do)

#	Question	Difficulty	Working Space
H6	A ribbon is $1\frac{1}{4}$ meter long and uses SEP . How many meters were used? $\frac{1}{3}$. How many meters were used?		
H7	One month of $\frac{1}{4}$ is sunny, the remaining $\frac{1}{3}$ is rainy, and the remaining $\frac{1}{2}$ is cloudy. What fraction of the month is cloudy? (Three levels of residual reasoning)		

H8	A water tank contains several liters of water. After using SEP , 18 liters remain. How many liters of water did the water tank originally contain? (reverse reasoning) $\frac{2}{5}$ Afterwards, 18 liters remain. How many liters of water did the water tank originally contain? (reverse reasoning)		
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VII. The Lessoncorecommon errorssummary

✔ Self-examination in this hall (tick after completion)

☐ I know the pitfalls of solving each knowledge point

☐ I can complete 🍌 basic questions independently

☐ I can challenge 🌿 advanced questions

☐ I remember the formula

🎯 Review of Learning Objectives - After completing this lesson you should be able to:

☐ Identify all trap types in our hall

☐ Solve 🍌 basic questions independently (100% correct)

☐ Challenge 🌿 Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	"What fraction is left" is directly multiplied by the original total amount	Calculate the true remaining amount first, then use the remaining amount $\times$ fraction. Baseline amounts may vary each time!
2	Skip intermediate calculations in multi-step problems	Calculate step by step, record "how much is used" and "how much is left" every time, without skipping steps.
3	"A's C/B" uses division incorrectly	"s" = multiplication: $A \times C/B$. Division is only used if "A is a fraction of B".
4	Wrong direction when reasoning backwards	Reduction formula: original quantity = remaining quantity $\div$ (1 – fraction used). Not a ride!
5	Calculate mixed numbers without converting them to improper fractions	For any multiplication or division involving mixed numbers, the first step = converting improper fractions.
6	Unit confusion – fractions vs actual quantities	"1/3" may be a fraction (without unit) or an actual quantity (with unit), please read the context clearly!
7	Missing sentences/missing units/missing calculations in application questions	Must write "Answer:..." + unit. Calculation: Amount used in all steps + final remaining amount = total amount.